

Meta-analysis reveals asymmetric reduction in the genetic diversity of introduced populations of exotic insects

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Abstract Factors promoting the invasion success of introduced populations have been receiving increased attention in studies of biological invasions. Previous reports have indicated that successful invasions may be attributable to reduced genetic diversity in the invasive species. However, there is large variation in the magnitude and direction of the impact of exotic species that have remained unexplained. Here, we present a structured meta-analysis of papers investigating the genetic diversity of native and introduced populations of exotic insects using nuclear microsatellites and mitochondrial DNA sequences. The results indicate that invasion by exotic insects had an overall reducing effect on the genetic diversity of the invading population, with nonzero effect sizes for the number of alleles (NA), observed heterozygosity (Ho), expected heterozygosity (He) and nucleotide diversity (Nd). However, when analyzing different orders (e.g., Lepidoptera, Hemiptera), the effect sizes of NA , Ho and Nd in Lepidoptera were found to bracket zero, as did the effect size of He in Hemiptera. These results suggest an asymmetric reduction in the genetic

diversity of introduced populations of exotic insects, indicating diverse mechanisms underlying their successful invasion.

Keywords Biological invasion · Genetic diversity · Insects · Effect size · Meta-analysis

Introduction

Biological invasion is often described as a process by which organisms (such as plants, insects and birds) spread to new regions where they were previously absent to obtain population growth and range expansion (Elton 1958). Exotic species are assumed to have a competitive advantage over native species during interspecific competition (ENREF_2). Studies have found that biological invasion is a major threat to the environment and biodiversity of the invaded region (Sakai et al. 2001). For example, Argentine ants have had significant effects on the invertebrate biodiversity of northern California (Human and Gordon 1997); *Eupatorium adenophorum* Spreng and *Eupatorium odoratum* L. (forest killer, local name banmara) are unpalatable, highly competitive and dramatically decrease the biodiversity of the invaded populations (Kunwar 2003). Many explanations have been offered to explain the competitive advantage of exotic species (Alpert 2006; Qin et al. 2013). Thus, biological invasion is able to cause major damage to the invaded region. It is estimated that the annual losses caused by

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biological invasions in China would correspond to USD 14.45 billion, direct and indirect economic losses respectively accounting for 16.59 and 83.41 % of the total economic losses (Xu et al. 2006).

Adaptive evolution may occur in small or sparse populations and provide a means of mitigating or avoiding inverse density-dependent effects (i.e., Allee effects) (Kanarek and Webb 2010). For instance, argentine ants display reduced intraspecific aggression, and the concomitant abandonment of territorial behavior unique to introduced populations contributes to the elevated population densities that are directly responsible for its widespread success as an invader (Holway et al. 1998). Genetic and evolutionary processes may be key features in determining whether invasive species become established and spread (Sakai et al. 2001)_ENREF_10. It is said, for instance, that the amount, types and organization of genetic diversity within introduced populations will largely determine the capacity to respond to the local selection pressures imposed by the physical and biotic environment (Barrett 1992). Thus, the patterns of genetic variation in invasive populations are critically important for scientific and applied significances. Moreover, scientists have indicated that understanding how genetic diversity is distributed can help predict the potential for populations of invasive species to evolve (Sakai et al. 2001)_ENREF_11_ENREF_11. Upon reviewing studies that measured nuclear molecular diversity within introduced and source populations (representing 80 species of animals, plants, and fungi), significant losses of both allelic richness and heterozygosity in the introduced populations were observed, and large gains in diversity were rare (Dlugosch and Parker 2008). However, it has sometimes been reported that the resulting genetic diversity within introduced populations from multiple source populations can exceed the original genetic diversity occurring within each native population (Facon et al. 2006).

Fortunately, several useful molecular marker tools to identify the genetic diversity of native and invasive populations of exotic species have been developed. Microsatellites are one type of molecular marker that has been used widely in recent years, and researchers have increasingly used microsatellites to identify the genetic diversity of introduced populations (Taylor and Cooper 1999). Mitochondrial DNA (mtDNA) sequencing is also widely used to characterize the population genetic

structures of invasive species and to carry out population assignment tests (Margaritopoulos et al. 2007; Ross and Shoemaker 2008) in invasive insects.

Although many studies have investigated into the genetic diversity of exotic insects using microsatellites (Henshaw et al. 2005; Ross et al. 2007; Vargo 2003a; Vargo et al. 2006) and mtDNA, quantitative data of whether a decrease in genetic diversity of exotic insects will aid the invasive process has remained elusive, and no mechanisms that might underlie this phenomenon have ever been satisfactorily established. However, as many studies have examined the genetic diversity of native and introduced populations of exotic insects by microsatellites (Fournier et al. 2008; Keever et al. 2013; Perdereau et al. 2013) and mtDNA (Downie 2002), a meta-analysis of these studies offers a unique chance to solve this problem based on statistical comparison and synthesizing the results of independent studies that have tested the same hypothesis (Cooper and Hedges 1994). By estimating the magnitude of the effect and analyzing the sources of variation, meta-analysis provides the perfect tool to review the results of previous studies. Meta-analysis has been used frequently to test ecological hypotheses (Stewart 2010). Here, we collected genetic diversity data acquired using microsatellites and mtDNA sequencing in invasive insects and applied a meta-analysis to summarize the effects of biological invasion on the genetic diversity of exotic insects.

Methods

Literature search

A list of invasive insects was collected from Wikipedia (http://en.wikipedia.org/wiki/Lists_of_invasive_species). Invasive insects span the globe. A literature search of peer-reviewed publications published between January 1980 and December 2014 reporting on investigation results on the genetic diversity of invasive insects was performed using the ISI-Web of Science research database. Papers that have studied the genetic diversity of native and introduced populations of invasive insects using microsatellites or mitochondrial DNA sequences were selected by using the following search terms:

<Name of invasive insect > \$AND microsatellite or Name of invasive insects \$AND mitochondria DNA.

For papers investigating the genetic diversity of invasive insects with microsatellites and mitochondrial DNA sequences, the number of alleles (*NA*), observed heterozygosity (*Ho*) and expected heterozygosity (*He*) in microsatellites and nucleotide diversity (*Nd*) in mitochondrial DNA sequences were selected to estimate the genetic diversity of native and introduced populations of invasive insects. Studies that evaluated *NA*, *Ho*, *He*, or *Nd* in both native and invasive populations were selected and obtained for the meta-analysis.

Data collection

For each observation within each study, we collected the means of the control treatment (that is, a native population of the invasive insect) and that of the experimental treatment (that is, an invasive population of the invasive insect), as well as their standard deviations (SD) and replicate numbers (*n*). For studies that did not report means and SDs, we calculated the average coefficient of variation (CV) for the control treatment and experimental treatment and then approximated the missing SD by multiplying the reported mean by the average CV.

Meta-analysis

The unit of analysis was a pair of invaded (*i*) and non-invaded sites (*n*). For each case, *Hedges' d* was used to estimate the effect size because (1) it weighs cases by their number of replications and the inverse of their variance and (2) it accounts for the effects of small sample sizes (Rosenberg et al. 2000). *Hedges' d* was calculated as follows:

$$d = \frac{\bar{X}^n - \bar{X}^i}{S} J \quad (1)$$

where *S* is the pooled standard deviation of *NA*, *Ho*, *He* and *Nd*. *S* was calculated as follows:

$$S = \sqrt{\frac{Si^2(Ni - 1) + Sn^2(Nn - 1)}{Ni + Nn - 2}} \quad (2)$$

And *J* is a weighting factor based on the number of replicates (*n*) in each case. *J* was calculated as follows:

$$J = 1 - \frac{3}{4(Nn + Ni - 2) - 1} \quad (3)$$

where *Nⁿ* and *Nⁱ* are replicate numbers (*n*) of the non-invaded and invaded sites, respectively.

The variance of *Hedges' d* was computed as follows:

$$Vd = \frac{Nn + Ni}{NnNi} + \frac{d^2}{2(Nn + Ni)} \quad (4)$$

Then, a weighted average effect size *d⁺* for a fixed effect model and the homogeneity test statistic *Q* were calculated as in Cooper and Hedges (1994). When *Q* was significant, the effect sizes were heterogeneous and differed among the included studies. In this situation, the random effects model (Cooper and Hedges 1994) to estimate effect sizes was used to analyze the related data. Accounting for the fact that, in addition to the sampling error, there is a random component of variation in effect sizes among studies, a random effects model was preferred. If the bias-corrected 95 % confidence interval (CI) of *d⁺* did not bracket zero, the effect was considered significant (Rosenberg et al. 2000). For the effect to be considered general, the homogeneity statistic *Q* should be insignificant for the studies included in the analysis. When the effect sizes of the studies were heterogeneous, we separated the studies into various organism groups according to classification status. All analyses were performed with R statistical program (Team 2013).

Results

Data collection and assessment

Our data collection covered 58 exotic invasive insects that were native to Asia, America, and Europe, among others. Hymenoptera, Lepidoptera and Coleoptera were the top three groups of insects with 19, 14 and 10 invasive species, respectively (Supplementary Table 1). By searching the literature published between January 1980 and December 2014 on these invasive insects, 10 invasive insects among Diptera, Hymenoptera, Isoptera, and Lepidoptera were found to include means, standard deviations and replicate numbers for *NA* in native and invaded populations published in 11 papers (Supplementary Table 2); 11

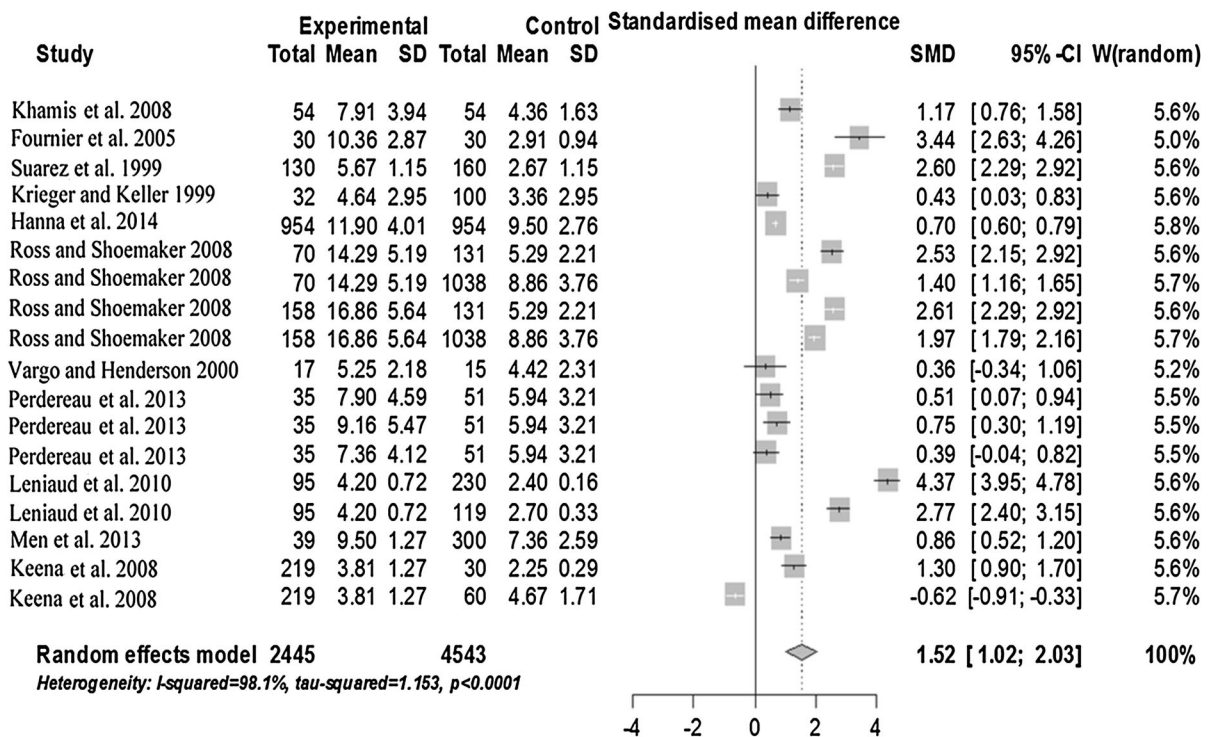


Fig. 1 Mean effect size (Hedges' d) of the impact of insect invasion on the overall NA. Experimental: introduced population; Control: native population; SMD standardized mean difference, 95 %-CI 95 % confidence interval, SD standard deviation

invasive insects among Coleoptera, Diptera, Hemiptera, Hymenoptera, Isoptera, and Lepidoptera included means, standard deviations and replicate numbers for *Ho* in native and invaded populations from 12 papers (Supplementary Table 3); 14 invasive insects among Coleoptera, Diptera, Hemiptera, Hymenoptera, Isoptera and Lepidoptera were found with means, standard deviations and replicate numbers for *He* in native and invaded populations from 14 papers (Supplementary Table 4); and 7 invasive insects among Lepidoptera, Isoptera, Hemiptera, Coleoptera and Diptera were found with means, standard deviations and replicate numbers for *Nd* in native and invaded populations from 7 papers (Supplementary Table 5).

Comparison of NA in native and introduced populations

The average effect size (Hedges' d) for NA in the present studies was 1.52 with a 95 % confidence

interval of 1.02–2.03, indicating an average effect size that was clearly not 0. These results indicated that NA was lower for invasive insects among invading populations than among native populations. Considering the qualitative data in all these studies, a decreasing effect of invasion on NA was found in fifteen out of eighteen studies (83.33 %) (Fig. 1). However, the effect sizes of the 18 studies were heterogeneous ($Q = 904.79$, $df = 17$, $p < 0.001$). We therefore separated the studies according to order: Hymenoptera, Isoptera and Lepidoptera. We found decreasing effects of invasion on NA in all orders except Lepidoptera; in Hymenoptera, the 95 % confidence interval was 1.26–2.60, and the average effect size was 1.93 ($Q = 424.18$, $df = 7$, $p < 0.001$, Fig. 2a); in Isoptera, the 95 % confidence interval was 0.15–2.91, and the average effect size was 1.53 ($Q = 285.05$, $df = 5$, $p < 0.001$, Fig. 2b); and in Lepidoptera, the 95 % confidence interval was -0.68–1.69, and the average effect size was 0.51 ($Q = 73.18$, $df = 2$, $p < 0.001$, Fig. 2c).

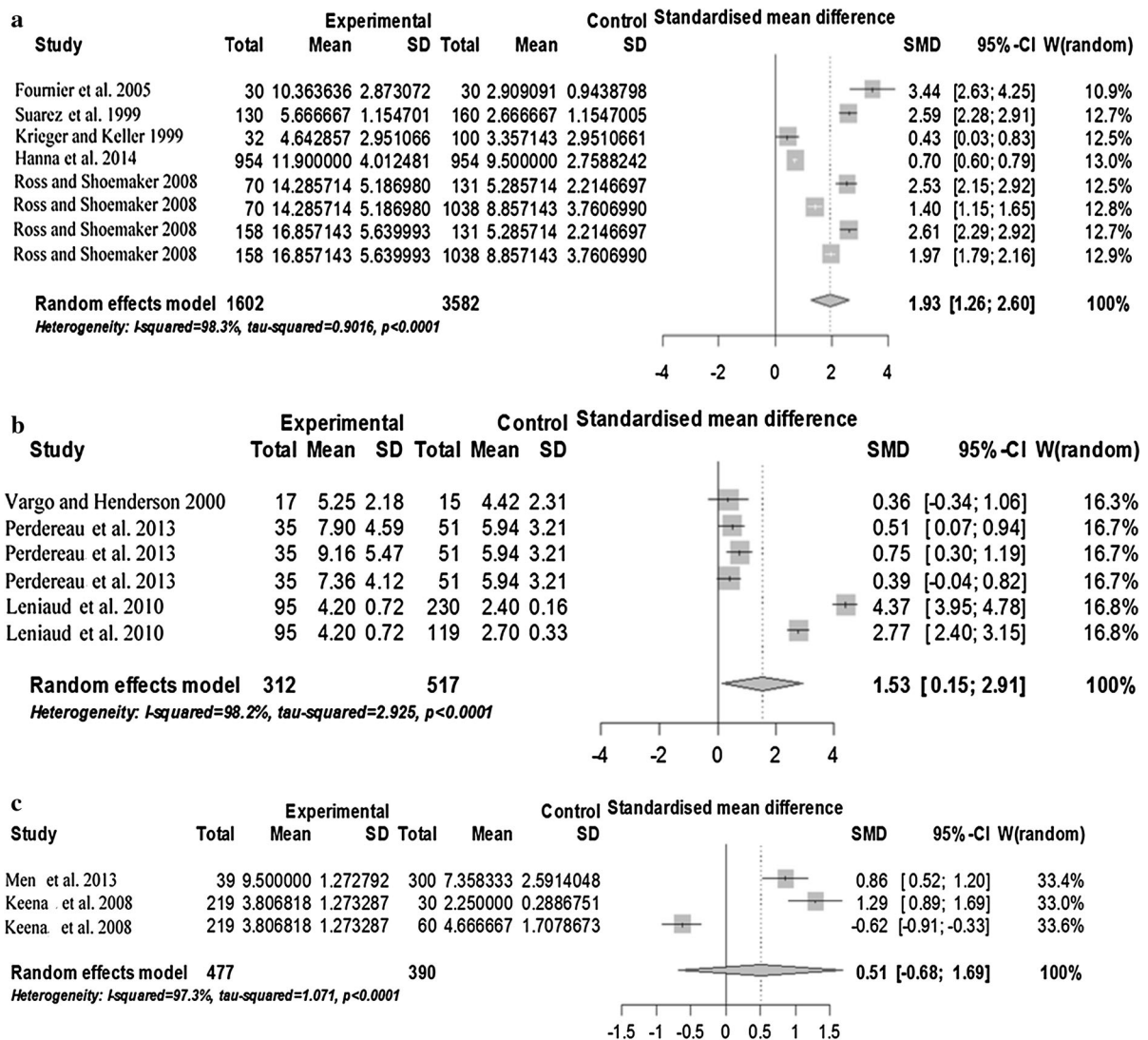


Fig. 2 Mean effect size (Hedges' d) of the impact of insect invasion on NA in various orders (**a** Hymenoptera; **b** Isoptera; **c** Lepidoptera). Experimental: introduced population; Control:

native population; SMD standardized mean difference, 95 %-CI 95 % confidence interval, SD standard deviation

Comparison of H_o in native and introduced populations

Among the fifteen studies examined, the average effect size for H_o was 0.90, and the 95 % confidence interval was 0.45–1.36, indicating an average effect size that was clearly different from 0 and a decreasing effect assigned to H_o from the invasion of insects ($Q = 306.2$, $df = 14$, $p < 0.001$, Fig. 3). A negative effect of invasion on H_o was found in eleven of the fifteen studies (73.33 %) (Fig. 3). While comparing

the various orders, a decreasing effect of invasion on H_o was found in all orders except Lepidoptera. In Hymenoptera, the 95 % confidence interval was 0.09–2.30, and the average effect size was 1.19 ($Q = 104.25$, $df = 4$, $p < 0.001$) (Fig. 4a); in Isoptera, the 95 % confidence interval was 0.82–1.34, and the average effect size was 1.08 ($Q = 0.37$, $df = 2$, $p = 0.8324$) (Fig. 4b); in Hemiptera, the 95 % confidence interval was 0.47–1.16, and the average effect size was 0.82 ($Q = 2.01$, $df = 1$, $p = 0.1559$) (Fig. 4c); and in Lepidoptera, the 95 % confidence

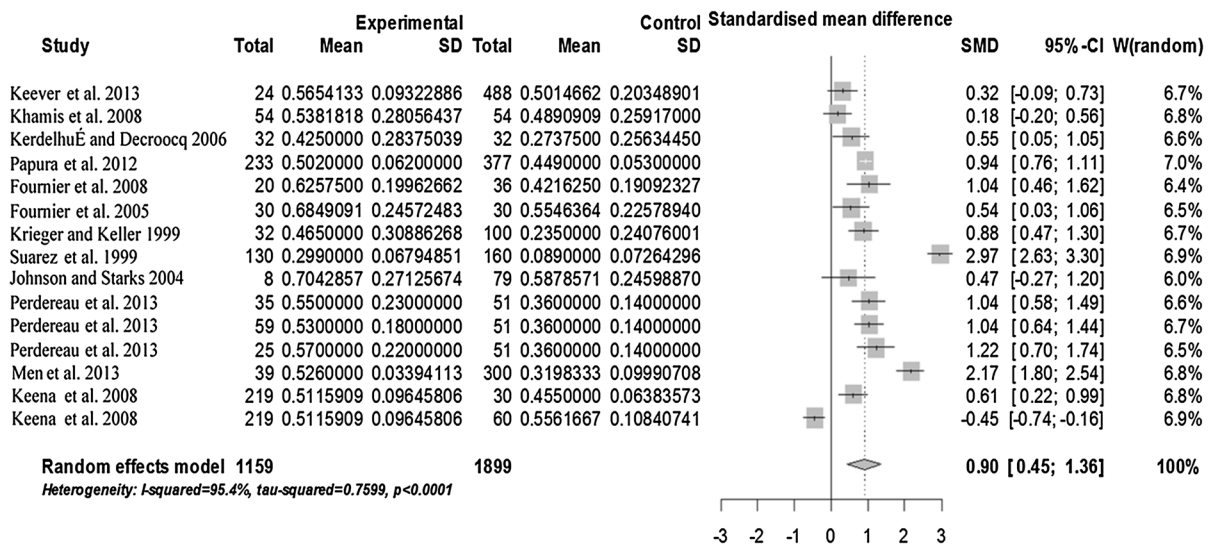


Fig. 3 Mean effect size (Hedges' d) of the impact of insect invasion on the overall H_o . Experimental: introduced population; Control: native population; SMD standardized mean difference, 95 %-CI 95 % confidence interval, SD standard deviation

interval was -0.77 – 2.32 , and the average effect size was 0.77 ($Q = 119.14$, $df = 2$, $p < 0.001$) (Fig. 4d).

Comparison of H_e in native and introduced populations

For H_e analysis, eighteen studies were extracted with data from native and invaded populations. Among these eighteen studies, the average effect size for H_e was 1.1065 , and the 95 % confidence interval was 0.5998 – 1.6132 , indicating that a decreasing effect was assigned to H_e by the invasion of insects ($Q = 845.57$, $df = 17$, $p < 0.001$, Fig. 5). Among the orders examined, Hymenoptera, Lepidoptera and Isoptera had 95 % confidence intervals that were clearly different from 0. In Hymenoptera, the average effect size for H_e was 1.64 , and the 95 % confidence interval was 0.58 – 2.70 ($Q = 231.6$, $df = 5$, $p < 0.001$, Fig. 6a); in Lepidoptera, the average effect size for H_e was 2.17 , and the 95 % confidence interval was 1.61 – 2.74 ($Q = 7.28$, $df = 1$, $p = 0.007$, Fig. 6b); and in Isoptera, the average effect size for H_e was 0.25 , and the 95 % confidence interval was 0.06 – 0.44 ($Q = 8.07$, $df = 5$, $p = 0.1522$, Fig. 6c). However, in Hemiptera, the 95 % confidence interval bracketed zero with an average effect size for H_e of 1.92 and a 95 % confidence interval of -0.37 – 4.21 ($Q = 66.64$, $df = 1$, $p < 0.001$, Fig. 6d).

Comparison of N_d in native and introduced populations

Among the fourteen studies in this group, the average effect size for N_d was 1.79 , and the 95 % confidence interval was 0.72 – 2.86 , indicating that a decreasing effect was assigned to N_d by the invasion of insects ($Q = 1162.6$, $df = 13$, $p < 0.001$, Fig. 7). However, among the various orders, only Coleoptera and Hemiptera had 95 % confidence intervals that were clearly different from 0 (Coleoptera: $Q = 2.46$, $df = 1$, $p = 0.1169$; Hemiptera: $Q = 310.74$, $df = 4$, $p < 0.001$, Fig. 8a, b), and the 95 % confidence interval for Lepidoptera bracketed zero ($Q = 474.2$, $df = 4$, $p < 0.001$, Fig. 8c).

Discussion

There are certain expressed phenotypes in introduced populations which undoubtedly facilitate their survival, reproduction and range expansion and further increase their probability of invasion success (Facon et al. 2006; Holway et al. 1998; Qin et al. 2013). However, a “genetic bottleneck” that results in lower levels of genetic diversity in introduced populations relative to native populations has been found in some studies (Keever et al. 2013; Tsutsui et al. 2000; Uller

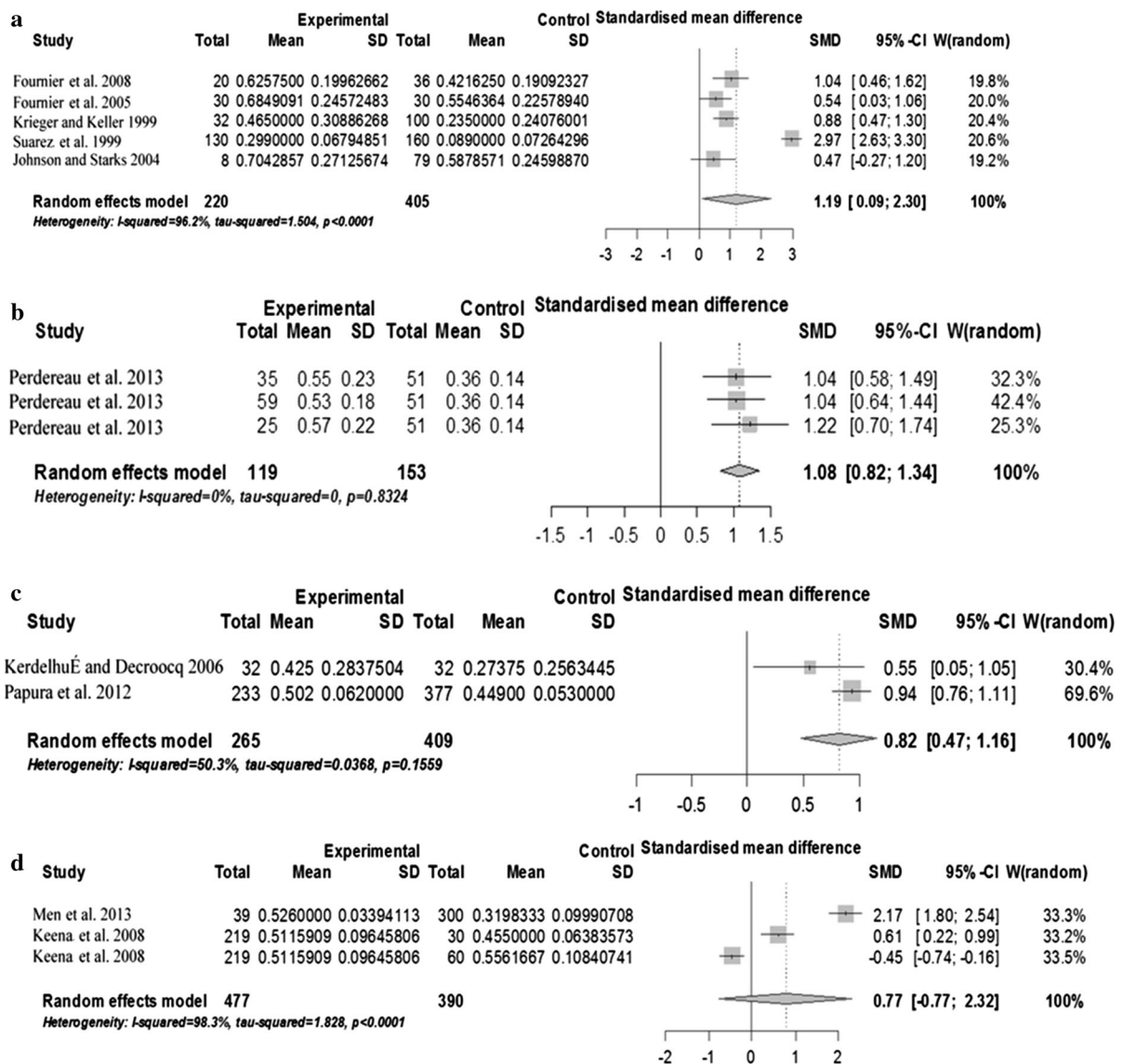


Fig. 4 Mean effect size (Hedges' d) of the impact of insect invasion on H_o in various orders (**a** Hymenoptera; **b** Isoptera; **c** Hemiptera; **d** Lepidoptera). Experimental: introduced

population; Control: native population; SMD standardized mean difference, 95 %-CI 95 % confidence interval, SD standard deviation

and Leimu 2011), and controversial examples have been reported in other studies (Facon et al. 2006). Overall, our meta-analysis showed that successful biological invasion by an insect tended to reduce the genetic diversity of that insect. This result has been found in several studies (Cai et al. 2008; Tsutsui et al. 2000). Generally, the relationship between the success of an invasive population and the genetic changes can be observed during the invasion process, and involves genetic drift, population admixture and/or natural

selection (Facon et al. 2006). Moreover, the genetic diversity of an invasive population is considered to be an important influence on its survival and adaptive capacity (Petit et al. 2004; Sakai et al. 2001)_ENREF_31. Thus, the suggestion that biological invasion decreases genetic diversity is supported by our meta-analysis. The overall trend toward a lower genetic diversity level following a biological invasion agrees with previous reviews (Branchiccela et al. 2014; Jones and Brown 2014; Yu et al. 2014) and is

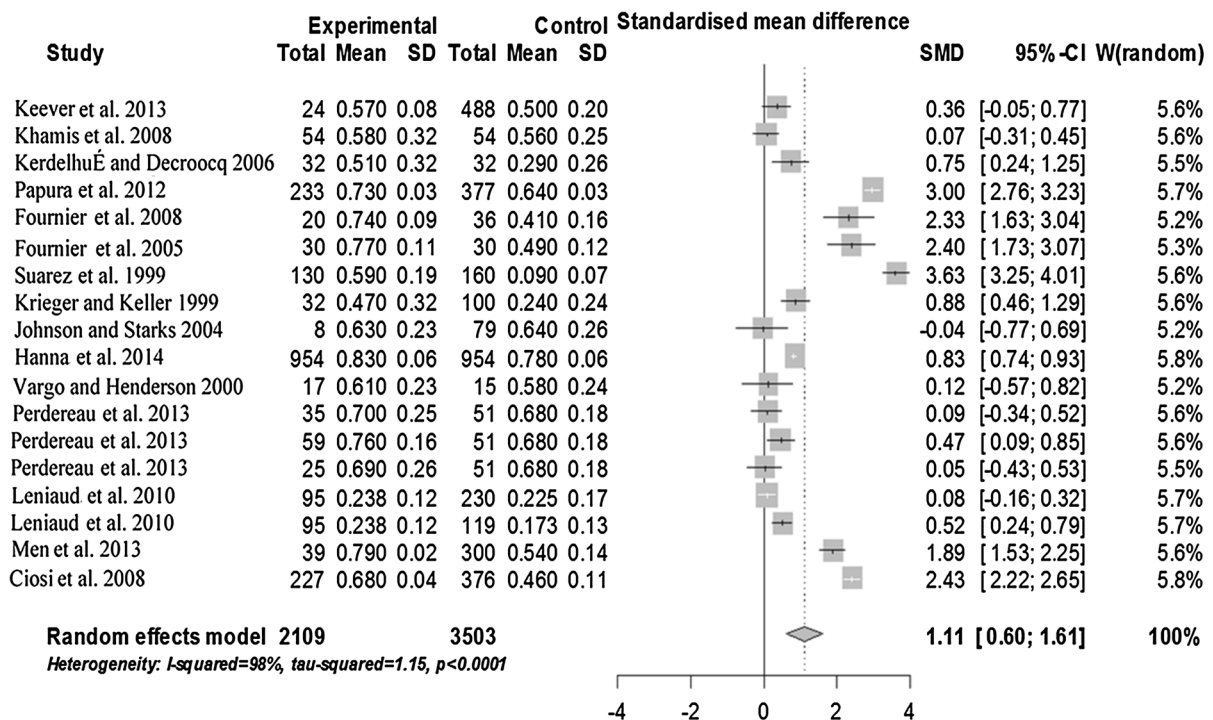


Fig. 5 Mean effect size (Hedges' d) of the impact of insect invasion on the overall He . Experimental: introduced population; Control: native population; SMD standardized mean difference, 95 %-CI 95 % confidence interval, SD standard deviation

consistent with the hypothesis that invasion reduces genetic diversity. The results indicate that the genetic diversity of Hymenopteran insects is lower in invasive populations than in native populations (Figs. 2a, 4a, 6a), consistent with reports that invading insects often undergo a genetic bottleneck (Mok et al. 2014; Omondi et al. 2014; Tsuchida et al. 2014) and therefore have a reduced level of genetic diversity. In the well-studied example of fire ants and the Argentine ant *Linepithema humile*, the introduced populations, which fail to show intraspecific aggression, form geographically large supercolonies (Holl-dobler et al. 1980; Keller 1996; Keller and Passera 1989; Wilson 1971). Moreover, the effectiveness of a recognition system may be further compromised by a loss of genetic variation. For example, a genetic bottleneck following introduction and establishment would result in genetic homogenization across spatially separate nests, which could further impair recognition (Suarez et al. 1999). Thus, a decrease in genetic diversity in Hymenopteran insects may be a key factor aiding the invasion process. Detecting

bottlenecks in recent invasions is a common component of attempting to assess genetic structure of invasive species (Fitzpatrick et al. 2012). However, very recent bottlenecks are difficult to detect because the expected deviation from mutation-drift equilibrium appears after several generations of reduced population size (Fitzpatrick et al. 2012). Thus to determine the relative importance of population size to invasiveness was justified as a way to testing bottlenecks (Alexander et al. 2009).

Similarly to the findings for Hymenoptera, our results suggest that low genetic diversity in introduced populations may facilitate the invasion success of Isopteran insects. Compared with native termite populations, introduced populations usually develop certain characteristics or factors that might be associated with their invasion success (Perdereau et al. 2011). In the subterranean pest termite *Reticulitermes flavipes*, introduced populations have often been identified with a high proportion of extended families following the production of a large number of secondary reproductives called neotenic

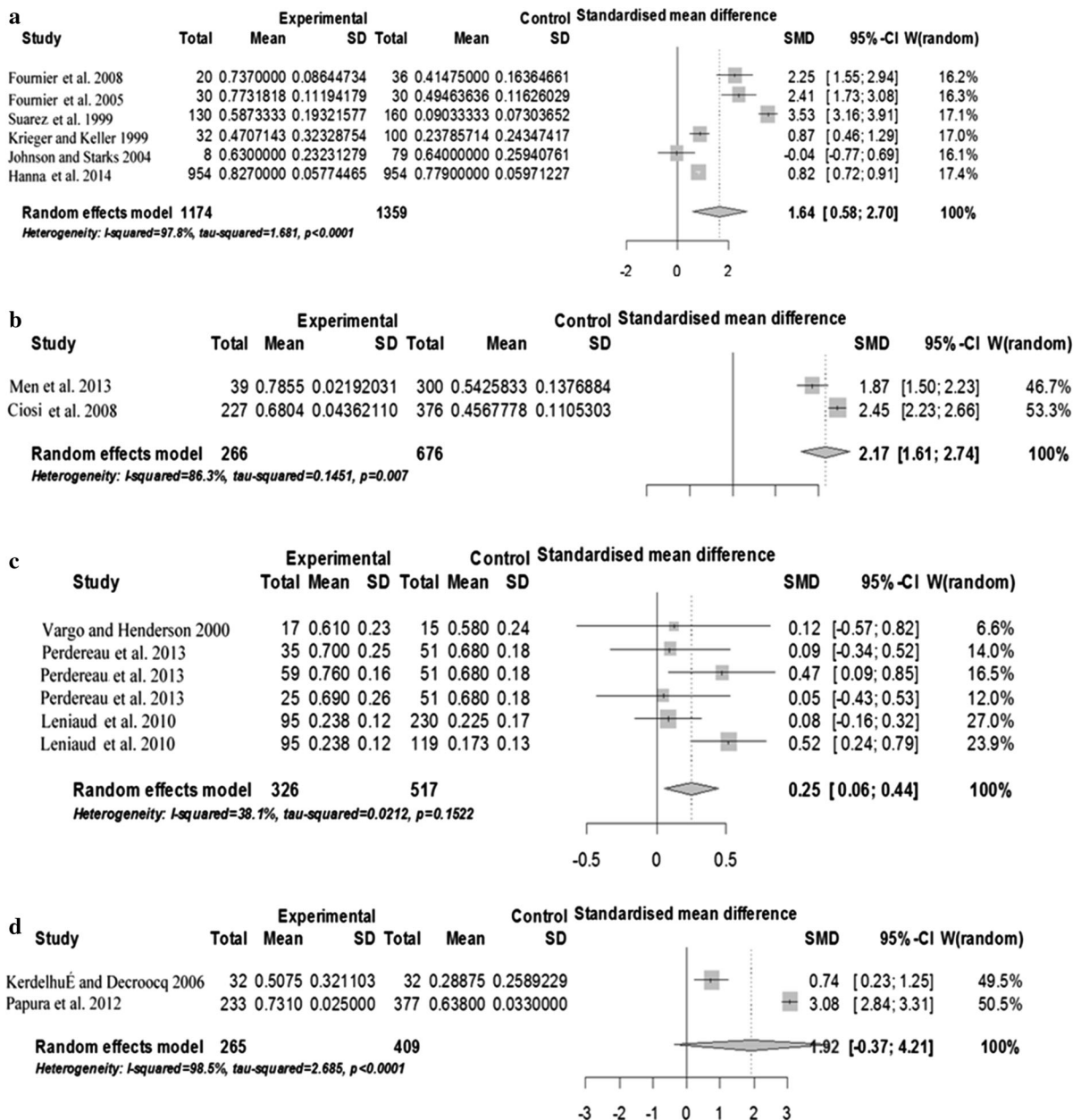


Fig. 6 Mean effect size (Hedges' d) of the impact of insect invasion on He in various orders (**a** Hymenoptera; **b** Isoptera; **c** Hemiptera; **d** Lepidoptera). Experimental: introduced

population; Control: native population; SMD standardized mean difference; 95 %-CI 95 % confidence interval, SD standard deviation

or mixed families by colony fusion (DeHeer and Vargo 2008; Vargo 2003b)_ENREF_45. Numerous reproductives in extended or mixed families are comparable to the supercolonies or unicolonies formed by invasive ants, in which individuals can mix freely among nests (Hölldobler and Wilson

1990). This type of social organization can contribute to high density and successful interspecific competition, further resulting in a lack of aggression between colonies, homogeneity in chemical recognition cues and reduced genetic diversity (Perdereau et al. 2010, 2011, 2013).

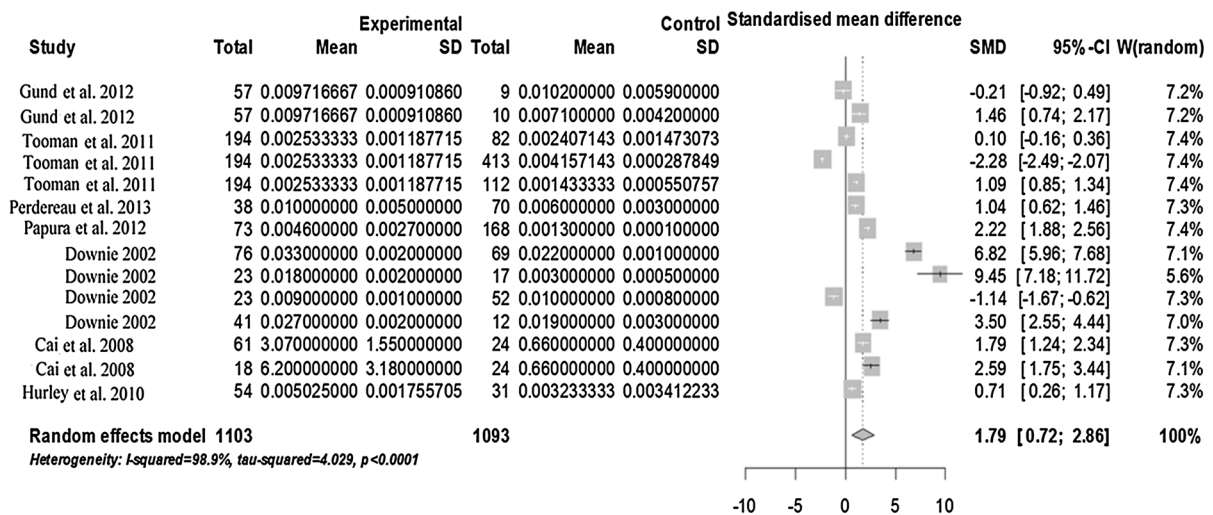


Fig. 7 Mean effect size (Hedges' d) of the impact of insect invasion on the overall N_d . Experimental: introduced population; Control: native population, SMD standardized mean difference; 95 %-CI 95 % confidence interval, SD standard deviation

However, in Lepidoptera, following the invasion we did not detect any decreasing effect on NA , Ho or N_d (Figs. 2c, 4d, 8c), while in Hemiptera, the 95 % confidence interval brackets zero for He (Fig. 6d).

Researchers have noted that introduction processes can be classified into three categories according to their expected influence on genetic diversity: (i) contiguous expansion after an irreversible event of natural barrier removal (e.g., canal opening); (ii) several discrete introduction events from several native populations; and (iii) a single introduction event from a single native population (Puillandre et al. 2008). Researchers have found that contiguous expansion occurs after an irreversible event of natural barrier removal. In this situation, the number of colonizing individuals is limited only by the dispersal capacity of the species (Hassan et al. 2003), and the genetic diversity of the introduced population may be similar to that of the original population. Another case involves several discrete introduction events from several native populations, with the mixture of these populations resulting in a high level of diversity in the introduced population (Genton et al. 2005). Some insects with wings, especially Lepidoptera, migrate long distances; thus, during the invasion process, the population of the native region can be introduced to an invaded region continuously. As a result, higher genetic diversity is often observed in the populations of these insects. In fact, we might expect markedly

different invasion patterns depending on the nature of the invasive insect and that of the invaded population.

Other factors, such as the reproductive system, genetic drift, natural selection, gene flow and human disturbance, also contribute to genetic diversity in exotic insects (Charlesworth 1996; Kuehn et al. 2003). In the medfly *Ceratitis capitata*, genetic drift, bottleneck effects, selection, and gene flow can lead to considerable genetic variation in populations from various geographic regions (Gaspero et al. 2002). In fact, in this situation, we were unable to find any negative effect on genetic diversity due to biological invasion. Thus, we conclude that a negative effect of biological invasion on genetic diversity can be expected in common situations, but not necessarily in the specific situation mentioned above.

Because human activity is becoming more and more widespread, many species have been introduced and expanded into new regions as a result of global trade and human transport. Biological invasions have caused serious damage, both ecologically and economically. Thus, it is critical to better understand how an exotic species becomes a successful invader. In recent years, more and more studies have suggested using microsatellites or mitochondrial DNA to investigate this problem (Hanna et al. 2014; Keever et al. 2013; Suarez et al. 1999). However, in our meta-analysis, we found that investigating the problem using only microsatellites or mitochondrial DNA may

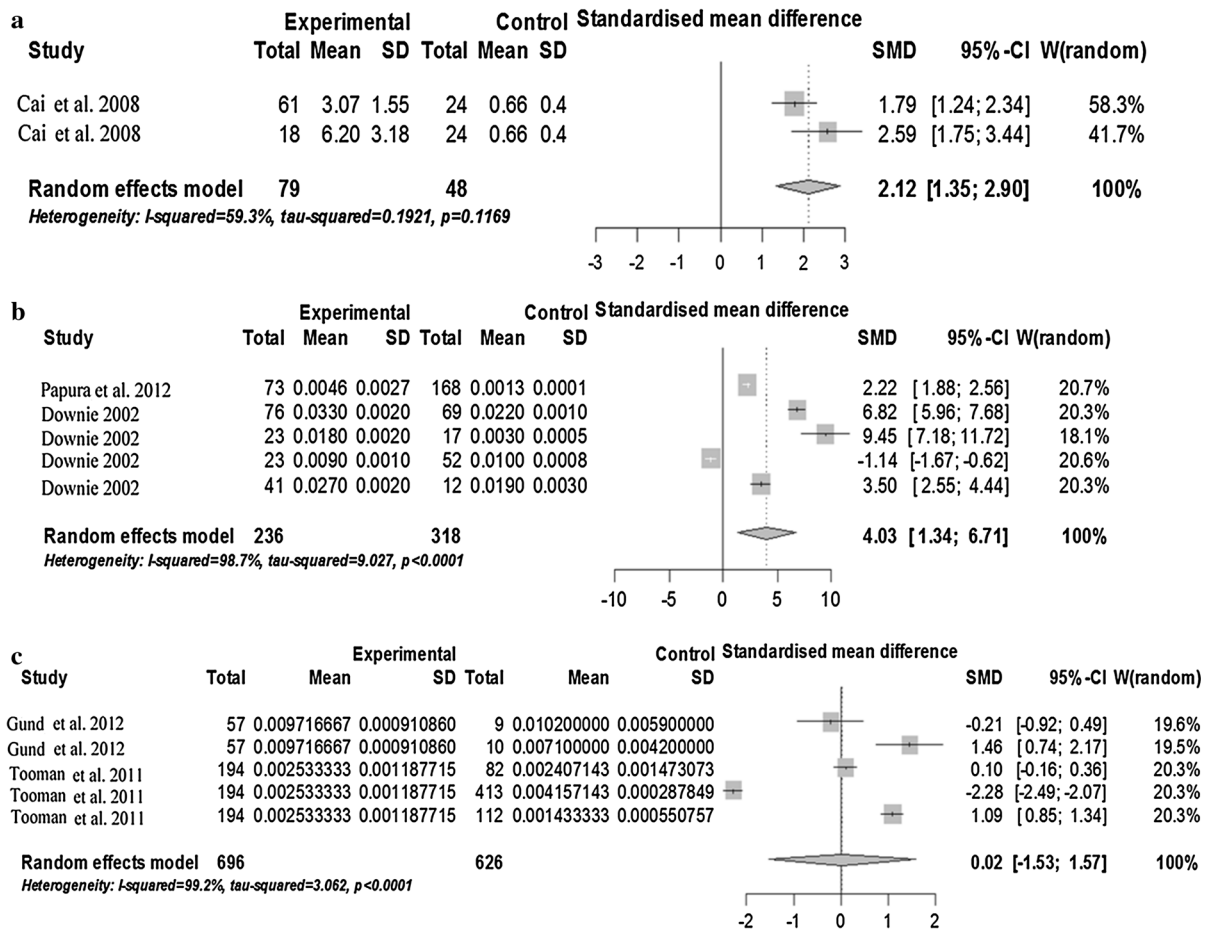


Fig. 8 Mean effect size (Hedges' d) of the impact of insect invasion on N_d in various orders (**a** Hymenoptera; **b** Isoptera; **c** Lepidoptera). Experimental: introduced population; Control:

native population; SMD standardized mean difference; 95 %-CI 95 % confidence interval, SD standard deviation

lead to negative results. Thus, we strongly recommend applying more than one method when investigating the genetic diversity and invasion profiles of exotic insects.

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Author Contributions Conceived and designed the study: YJX, DFC. Collected and analyzed the data and designed the simulations: DFC, YJX. Wrote the first draft of the manuscript: YJX, DFC, GWL.

Compliance with ethical standards

Conflict of interest None.

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